先进计算与数字工程研究中心

PATTERN RECOGNITION AND MACHINE LEARNING

Lecture 07 – Foundations of Statistical Learning Theory

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Training accuracy and next day failure

Night shift result

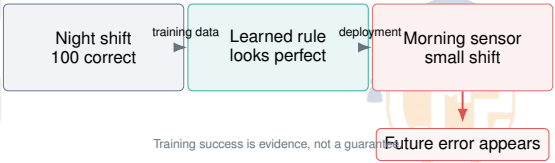
A classifier scores 100 percent on yesterday's inspection records.

Morning change

A new sensor batch shifts one measurement by a small amount.

Approval question

Does the perfect training score still justify deployment?



Six syllabus questions in three hours

Syllabus topic	Time	Question
Statistical learning framework	0.5 h	What is learned from what?
Empirical and expected risk	0.5 h	Which risk can we observe?
Test error estimation	0.5 h	Which evidence stays independent?
Regularization	0.5 h	How is capacity controlled?
Bias and variance	0.5 h	Why does the fitted rule change?
Learning theory	0.5 h	When does generalization become plausible?

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Separate learning, model selection, and final testing

- Training chooses model parameters.
- Validation chooses settings and compares candidate methods.
- Testing estimates performance after these choices are finished.
- The split must match future use: time, groups, or independent examples.

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The learning problem has five named parts

Part	Meaning
Input space $\mathcal{X}$	Possible observations
Output space $\mathcal{Y}$	Prediction targets
Unknown distribution P	How future pairs arise
Hypothesis space $\mathcal{H}$	Rules the learner may choose
Loss L	Cost of a prediction error

Inputs and targets belong to different spaces

Input

$x \in \mathcal{X}$ may contain measurements, text, images, or engineered features.

Target

$y \in \mathcal{Y}$ may be a real value, a class, or a structured object.

Type check

A probability, a score, and a class label are different outputs.

A sample is a window onto a population

Observed

$S = \{(x_i, y_i)\}_{i=1}^{N}$ is one finite sample.

Unobserved

Future pairs follow an unknown population distribution $P(x, y)$.

Gap

The sample can differ from the population by chance or by collection bias.

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The unknown distribution carries the future

$$(x_i, y_i) \sim P, \qquad (x, y)_{\text{future}} \sim P$$

Independent and identically distributed sampling

Independent

One observation does not reveal another observation's random error.

Identically distributed

All observations follow the same population mechanism.

Reasonable case

Randomly sampled products from a stable production line.

Broken case

Several visits from the same patient treated as unrelated people.

The hypothesis space defines what can be learned

Hypothesis space

$\mathcal{H}$ is the set of prediction rules available to the learner.

Linear family

$f_w(x) = w^o px + b$ for all allowed w and b.

Consequence

A pattern outside $\mathcal{H}$ cannot be recovered by better optimization alone.

A learning algorithm chooses within the space

$$A : S \longmapsto \hat{f}_S \in \mathcal{H}$$

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Loss turns an error into a number

Task	Loss	What it emphasizes
Classification	$\mathbf{1}(\hat{y} \neq y)$	Wrong decisions
Regression	$(\hat{y} - y)^2$	Large residuals
Robust regression	$\lvert\hat{y} - y\rvert$	Typical absolute error
Probability	$-\log p(y \mid x)$	Confident wrong probabilities

Zero one loss counts wrong classifications

$$L(\hat{y}, y) = \begin{cases} 0, & \hat{y} = y \\ 1, & \hat{y} \neq y \end{cases}$$

Squared loss magnifies large residuals

$$L(\hat{y}, y) = (\hat{y} - y)^2$$

The prediction rule differs from the learning rule

Prediction rule

f maps an input to an output.

Learning rule

A maps a dataset to a prediction rule.

Common confusion

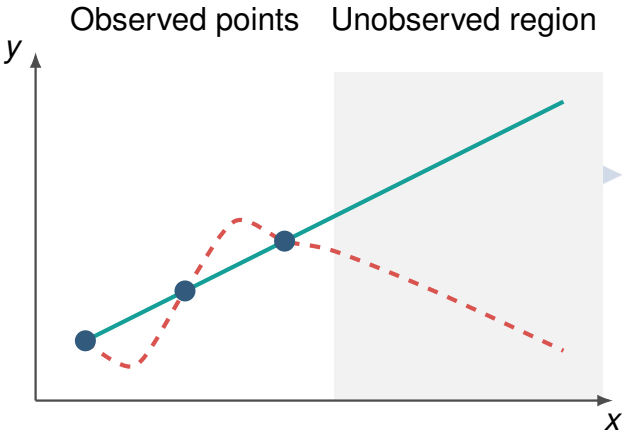
A linear predictor may be learned by many different procedures.

Parameters and hyperparameters play different roles

Quantity	Chosen by	Example
Parameter	Training optimization	Regression coefficient
Hyperparameter	Validation procedure	Penalty strength
Design choice	Problem formulation	Loss and feature set

Different rules can fit the same training points

- Training points do not determine one unique function.
- Smoothness, locality, and simplicity guide predictions between and beyond them.
- A useful preference must suit the task.



Both curves pass through all three training points.

No free lunch for model families

Claim

Averaged uniformly over every possible target rule, no learner dominates all others.

Practical meaning

Performance gains come from matching assumptions to a problem class.

Misreading

The theorem does not say that model choice is pointless.

A supermarket demand problem

Input

Price, promotion, weekday, weather, and store location.

Target

Tomorrow's unit sales for each store.

Population question

Will a holiday week follow the same distribution as ordinary weeks?

A medical screening problem

Decision

Refer a patient for an expensive confirmatory test.

Asymmetric loss

Missing a serious case costs more than one unnecessary referral.

Sampling

A specialist hospital may not represent community clinics.

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Expected risk measures future loss

$$R(f) = \mathbb{E}_{(x,y)\sim P}[L(f(x), y)]$$

Empirical risk averages observed loss

$$\widehat{R}_S(f) = \frac{1}{N} \sum_{i=1}^{N} L(f(x_i), y_i)$$

One sample gives one risk estimate

Fixed model

Different random samples produce different empirical risks.

Large sample

Averages usually fluctuate less when observations are representative.

Biased sample

More data from the wrong population do not repair the target mismatch.

Empirical risk minimization selects a fitted rule

$$\hat{f}_S \in \arg\min_{f \in \mathcal{H}} \widehat{R}_S(f)$$

Approximation error comes from the model family

Best in class

$f^*_{\mathcal{H}} \in \arg\min_{f \in \mathcal{H}} R(f)$

Approximation gap

$R(f^*_{\mathcal{H}}) - R(f^*)$

Cause

The hypothesis space cannot express the population optimum.

Estimation error comes from finite data

Quantity

$R(\hat{f}_S) - R(f^*_{\mathcal{H}})$

Cause

The sample makes one candidate look better than another by chance.

Dependence

A richer space can increase this uncertainty.

Optimization error comes from incomplete search

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Quantity

$\widehat{R}_S(\tilde{f}) - \min_{f \in \mathcal{H}} \widehat{R}_S(f)$

Cause

The numerical procedure stops early or reaches a poor solution.

Diagnosis

Low training loss suggests optimization is not the main problem.

Three gaps explain many failures

Gap	Question	Typical action
Approximation	Can the family express it?	Change representation
Estimation	Can finite data identify it?	More data or less capacity
Optimization	Did the solver find it?	Improve numerical search

Training error rewards memorization

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Observation

A larger hypothesis space cannot have higher minimum training error.

Reason

It contains every old candidate plus additional candidates.

Consequence

Training error alone always favors capacity.

Model selection creates a second level of fitting

Inner fitting

Estimate parameters for each candidate setting.

Outer fitting

Choose the candidate with the best validation score.

Implication

The validation set influences the final model even without gradient updates.

Trying many models increases lucky winners

Contest

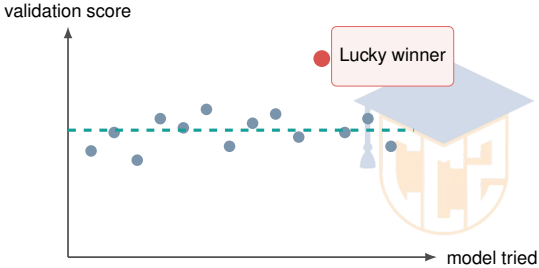
One hundred weak ideas are tested on the same validation set.

Winner

The best score reflects skill plus favorable validation noise.

Remedy

Use nested evaluation or confirm once on untouched data.



More tries create more opportunities for noise to win.

Leakage makes empirical risk dishonest

Leak	Example	Effect
Target leak	A post-outcome code	Future information enters features
Split leak	One patient in two folds	Related records cross the boundary
Preprocessing leak	Scaling before splitting	Validation statistics enter training

Noisy sine data and the generalization gap

Population signal

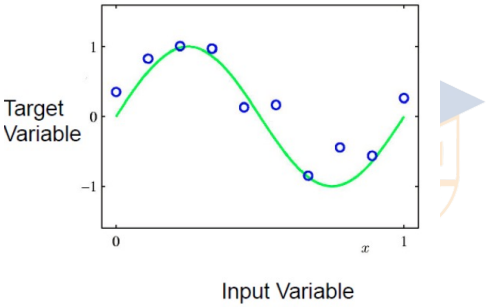
$y = \sin(2\pi x) + \epsilon$

Observed data

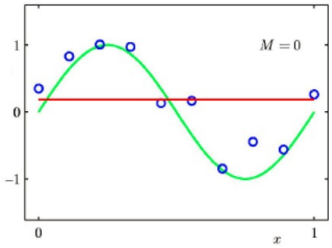
Only a small noisy sample is available.

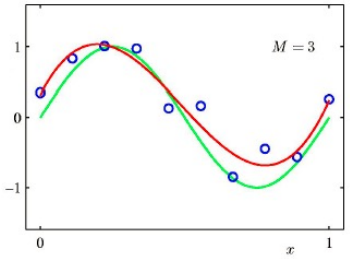
Challenge

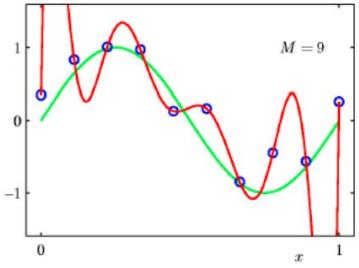
Many curves can pass close to the same points.



Underfitting and overfitting are different gaps







Underfitting. The family misses stable structure in both training and future data.

Useful fit. The model captures the signal without chasing every fluctuation.

Overfitting. The fitted rule follows sample noise that will not repeat.

Training and test curves move in opposite directions

Training

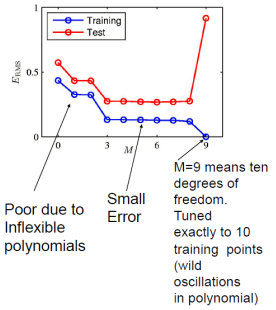
Error usually decreases as polynomial degree grows.

Test

Error often falls first and rises after variance dominates.

Selection

The best degree depends on independent validation, not the test set.



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More representative data can reduce estimation error

Same model family

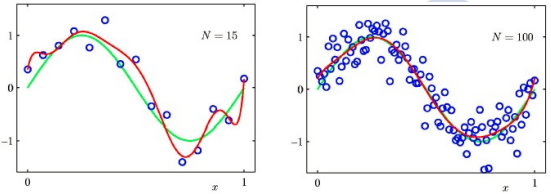
High degree remains available.

Larger sample

Random fluctuations become harder to fit without contradicting other points.

Limit

More biased data can reinforce the wrong population.



An independent test set estimates final performance

Purpose

Estimate the loss of the completed learning procedure on unseen data.

Requirement

No model choice may use test outcomes.

Scope

The estimate applies to the represented population and time period.

Training validation and test data have separate jobs

Split	Used for	Must not be used for
Training	Fit parameters	Final performance claim
Validation	Choose settings	Unbiased final claim
Test	Estimate final loss	Further selection

Holdout evaluation is one clean split

Procedure

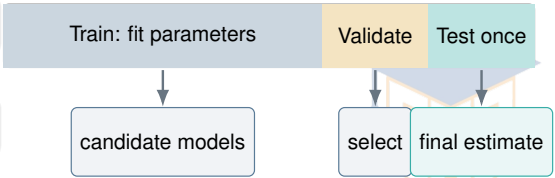
Fit on training data, choose on validation data, report on test data.

Strength

Simple and easy to audit.

Weakness

A small test set gives a noisy estimate.



The test set stays outside every design decision.

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Repeated test set reuse

Fresh test

Independent of every adaptive choice.

Spent test

Results have influenced features, thresholds, or model settings.

Repair

Collect a new test set or use a preregistered evaluation protocol.

Random splitting assumes exchangeable observations

Random split

Appropriate when observations can exchange positions without changing the problem.

Example

Independent products sampled from a stable line.

Failure

Time, groups, or spatial clusters create structure that random splitting breaks.

Stratified splitting preserves class proportions

Use

Classification with limited examples in some classes.

Benefit

Each split receives a similar class proportion.

Limit

Stratification does not repair group dependence or time leakage.

Time ordered evaluation predicts the future

Training

Use observations available before a cutoff.

Testing

Evaluate on a later period.

No shuffle

Random mixing lets future patterns inform the past.

Group splitting keeps related records together

Group

Patient, machine, household, speaker, or geographic site.

Rule

All records from one group stay in the same fold.

Reason

Otherwise the model can recognize the group rather than generalize.

Cross validation rotates the held out fold

Partition

Split the development data into K folds.

Repeat

Fit on $K - 1$ folds and evaluate on the remaining fold.

Aggregate

Average the K held out losses.

Fold					
1	validate	train	train	train	train
2	train	validate	train	train	train
3	train	train	validate	train	train
4	train	train	train	validate	train
5	train	train	train	train	validate

Every case is validated once and trained several times.

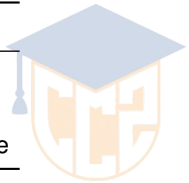
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The cross validation estimate averages held out loss

$$\widehat{R}_{\mathrm{CV}} = \frac{1}{K} \sum_{k=1}^{K} \widehat{R}_k$$

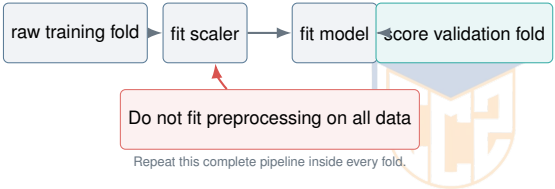
The number of folds changes the tradeoff

Choice	Training fraction	Computation	Typical behavior
Five folds	80 percent	Moderate	Often stable
Ten folds	90 percent	Higher	Less training size bias
Leave one out	Almost all	Very high	Can have high variance



Preprocessing inside cross validation folds

1. Fit scaling, imputation, or feature selection on the training folds.
2. Apply the fitted transformation to the held out fold.
3. Fit the model on transformed training folds.
4. Score only on the transformed held out fold.



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Nested cross validation separates selection and assessment

Inner loop

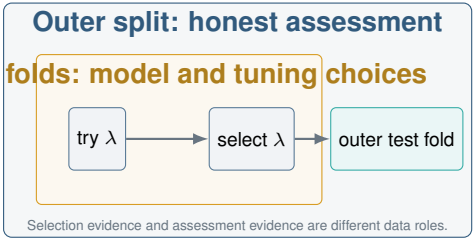
Choose hyperparameters using only inner training and validation folds.

Outer loop

Estimate the selected procedure on untouched outer folds.

Cost

Nested evaluation trains many models but protects the estimate.



Repeated hyperparameter search

Grid search

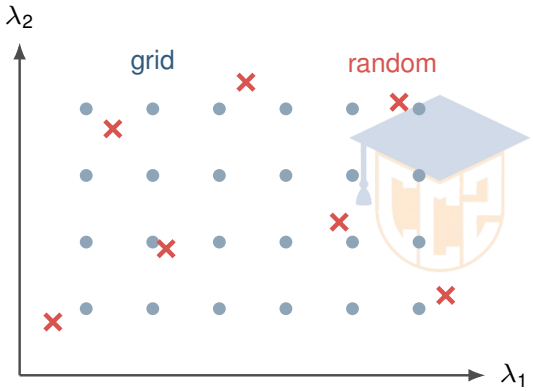
Evaluate every point in a predefined grid.

Random search

Sample combinations from specified distributions.

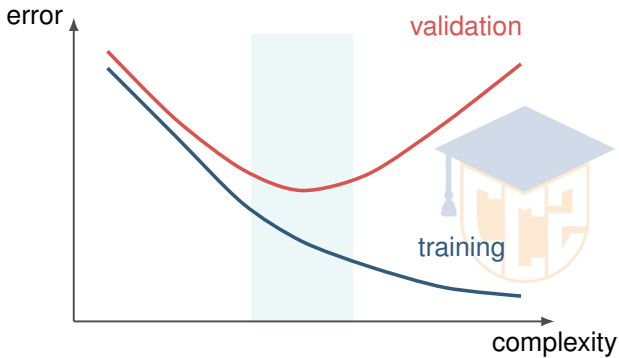
Shared rule

Both must stay inside the validation procedure.



Choose a stable region of validation performance

- Vary model complexity while keeping the split fixed.
- Training error can keep falling after validation error rises.
- Prefer a setting that also performs well in nearby settings and repeated splits.



Illustrative curves; compare repeated validation results.

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Cross validation in Python

```
from sklearn.datasets import load_breast_cancer
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import StratifiedKFold, cross_val_score
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler

X, y = load_breast_cancer(return_X_y=True)
pipe = make_pipeline(StandardScaler(), LogisticRegression(max_iter=3000))
folds = StratifiedKFold(n_splits=5, shuffle=True, random_state=7)
scores = cross_val_score(pipe, X, y, cv=folds, scoring='balanced_accuracy')
print(scores.round(3), scores.mean().round(3))
```

A point estimate needs uncertainty

Mean

Summarizes the center of repeated held out scores.

Spread

Shows how much the estimate changes across folds or resamples.

Caution

Fold scores share training data and are not fully independent replicates.

Learning curves diagnose data and capacity

Horizontal axis

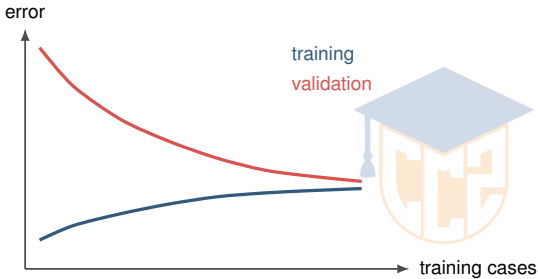
Number of training examples.

Curves

Training and validation loss as data grow.

Use

The gap and final level suggest different interventions.



Gap narrows as the sample grows.

Regularization controls effective capacity

Goal

Reduce sensitivity to sample details while retaining useful structure.

Methods

Penalty, constraint, early stopping, or restricted architecture.

Tradeoff

Excessive control can remove real signal.

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Penalized empirical risk

$$\hat{f}_\lambda = \arg\min_{f \in \mathcal{H}} \left[\widehat{R}_S(f) + \lambda \Omega(f) \right]$$

A penalty and a constraint can describe the same preference

$$\min_f \widehat{R}_S(f) + \lambda\Omega(f) \quad \Longleftrightarrow \quad \min_f \widehat{R}_S(f) \text{ s.t. } \Omega(f) \leq C$$

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The squared Euclidean norm shrinks coefficients smoothly

$$\Omega(w) = \|w\|_2^2 = \sum_{j=1}^{D} w_j^2$$

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The absolute-value norm can create sparse solutions

$$\Omega(w) = \|w\|_1 = \sum_{j=1}^{D} |w_j|$$

Norm geometry explains different solutions

Absolute-value norm ball

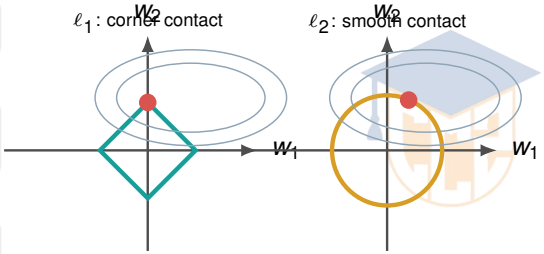
The one-norm corners lie on axes and encourage exact zeros.

Euclidean norm ball

The smooth two-norm boundary usually touches away from coordinate axes.

Scaling

The geometry changes meaning when feature units differ.



Feature scaling changes the meaning of a coefficient penalty

Meters

A distance coefficient may be small because the unit is large.

Centimeters

The same relationship needs a coefficient one hundred times smaller.

Consequence

Unscaled penalties prefer units, not necessarily simpler relationships.

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Coefficient growth reveals unstable flexibility

Observation

High degree polynomial coefficients can become extremely large.

Effect

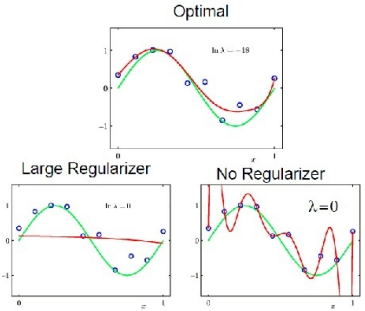
Small input changes create large output swings.

Response

Penalization trades a little fit for much greater stability.

	$M=0$	$M=1$	$M=6$	$M=9$
$w_0^\star$	0.19	0.82	0.31	0.35
$w_1^\star$		-1.27	7.99	232.37
$w_2^\star$			-25.43	-5321.83
$w_3^\star$			17.37	48568.31
$w_4^\star$				-231639.30
$w_5^\star$				640042.26
$w_6^\star$				-1061800.52
$w_7^\star$				1042400.18
$w_8^\star$				-557682.99
w_9				125201.43

Regularization changes the polynomial fit



Without penalty. A high degree curve follows noise and swings between points.

With penalty. The same basis produces a smoother curve with smaller coefficients.

Selection. Penalty strength still requires independent validation.

Early stopping is an implicit regularizer

Observation

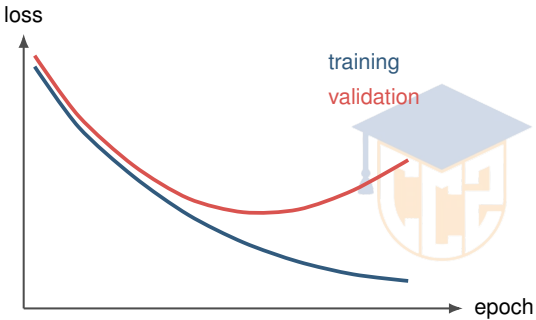
Iterative fitting often learns broad signal before fine sample details.

Rule

Stop when validation performance stops improving.

Protocol

The stopping epoch is a hyperparameter chosen by validation.



Stop near the minimum validation loss.

Regularization strength is a hyperparameter

Very small	Intermediate	Very large
Low training bias	Balanced fit	Strong shrinkage
Possible high variance	Often best validation	Possible underfitting

Regularization belongs inside model selection

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```
import numpy as np
from sklearn.datasets import load_diabetes
from sklearn.linear_model import Ridge
from sklearn.model_selection import KFold, cross_validate
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler

X, y = load_diabetes(return_X_y=True)
folds = KFold(n_splits=5, shuffle=True, random_state=3)
for alpha in [0.01, 0.1, 1.0, 10.0, 100.0]:
    model = make_pipeline(StandardScaler(), Ridge(alpha=alpha))
    out = cross_validate(model, X, y, cv=folds, scoring='neg_mean_squared_error')
    print(alpha, np.sqrt(-out['test_score']).mean().round(2))
```

The intercept usually escapes coefficient penalties

Reason

The intercept sets the baseline prediction rather than sensitivity to a feature.

Centered data

The intercept represents the mean target when features are centered.

Implementation

Check the library convention instead of assuming.

A Bayesian view turns penalties into priors

Gaussian prior

A zero centered Gaussian prior leads to a squared Euclidean coefficient penalty.

Laplace prior

A zero centered Laplace prior leads to an absolute-value coefficient penalty.

Interpretation

A prior is a modelling assumption, not a proof that coefficients are small.

Regularization does not repair data leakage

Can help

Reduce sensitivity to noisy training details.

Cannot help

Remove future information already embedded in a feature.

Priority

Fix the split and feature timestamp before tuning penalties.

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Capacity can be controlled without coefficient penalties

Method	Capacity control
Shallow tree	Limit depth or leaf size
Nearest neighbors	Increase neighborhood size
Feature map	Restrict basis complexity
Iterative model	Stop earlier

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Repeated datasets reveal model instability

Thought experiment

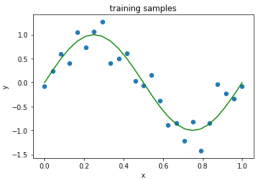
Draw many training sets from the same population.

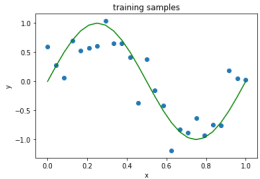
Fit

Run the same learning procedure on every set.

Observation

The predictions form a distribution, not one fixed answer.





The average fitted predictor

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$$\bar{f}(x) = \mathbb{E}_{\mathcal{S}}[\hat{f}_{\mathcal{S}}(x)]$$

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Bias measures systematic model mismatch

$$\text{Bias}(x) = \bar{f}(x) - f^*(x)$$

Variance measures sensitivity to the training set

$$\mathrm{Var}(x) = \mathbb{E}_S[(\hat{f}_S(x) - \bar{f}(x))^2]$$

Irreducible noise remains after perfect modelling

Data model

$y = f^*(x) + \epsilon$

Noise

$\mathbb{E}[\epsilon \mid x] = 0$ and $\mathrm{Var}(\epsilon \mid x) = \sigma^2$

Limit

No predictor using only x can remove unpredictable noise.

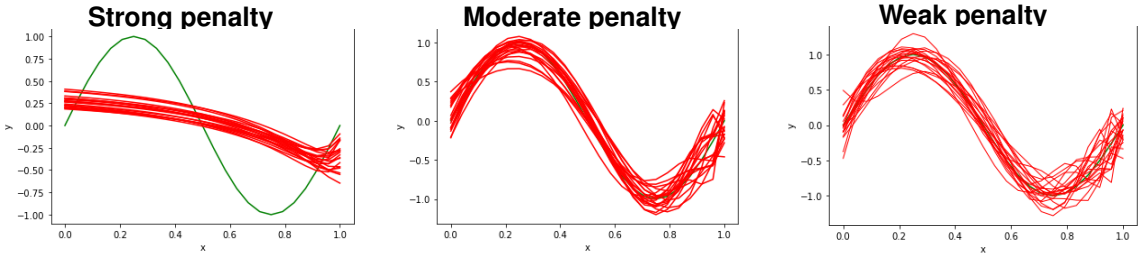
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Squared prediction error decomposes into three parts

$$\mathbb{E}_{S,y}[(\hat{f}_S(x) - y)^2] = \text{Bias}(x)^2 + \text{Var}(x) + \sigma^2$$

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Repeated fits make bias and variance visible



Stable but wrong. Curves cluster together away from the true signal.

Best region. Moderate shrinkage balances fit and stability for these samples.

Unstable. Curves spread widely across training samples.

Regularization moves bias and variance in opposite directions

Weak penalty

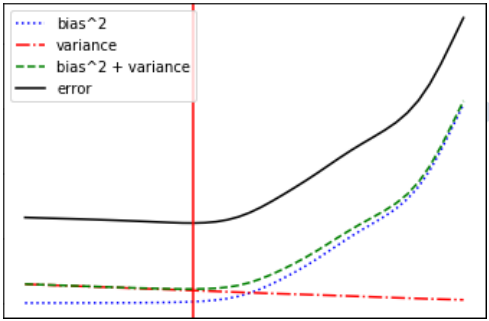
Low bias may come with high variance.

Moderate penalty

A small bias increase can buy a large variance reduction.

Strong penalty

Excessive shrinkage creates high bias.



Three model complexities across repeated samples

Simple

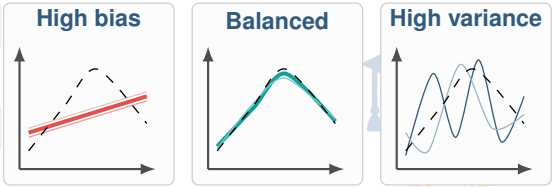
Similar fits with a stable systematic miss.

Balanced

Fits vary modestly around the signal.

Complex

Fits react strongly to the sampled noise.



Dashed line is the population signal.

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Training error and generalization error form a capacity curve

Low capacity

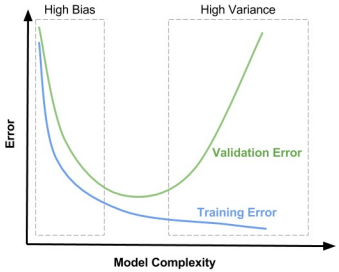
Both errors can be high.

Intermediate capacity

Generalization error reaches its minimum.

High capacity

Training error falls while the generalization gap grows.



Sample size changes effective complexity

Fixed family

The same model can have high variance with ten samples and low variance with ten thousand.

Reason

More observations constrain more degrees of freedom.

Comparison

Model complexity has meaning only relative to data and noise.

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High bias leaves two curves high and close

Training curve

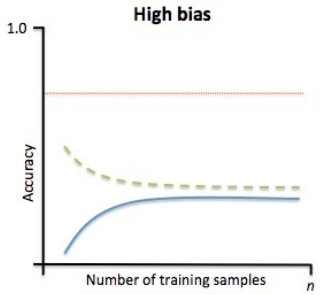
Settles at a high error.

Validation curve

Settles near the training curve.

Likely action

Improve representation or reduce excessive regularization.



High variance leaves a persistent gap

Training curve

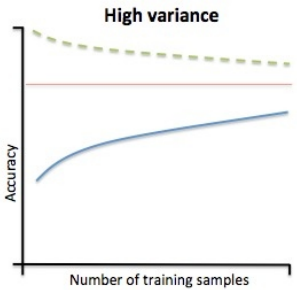
Remains much better than validation.

More data

The gap may narrow as training size grows.

Likely action

Collect representative data or strengthen capacity control.



A small simulation estimates prediction variance

```
import numpy as np
from sklearn.preprocessing import PolynomialFeatures
from sklearn.linear_model import LinearRegression
from sklearn.pipeline import make_pipeline

rng = np.random.default_rng(4)
x0 = np.array([[0.35]])
for degree in [1, 3, 9]:
    predictions = []
    for _ in range(100):
        X = rng.uniform(0, 1, size=(25, 1))
        y = np.sin(2*np.pi*X[:, 0]) + rng.normal(0, 0.2, 25)
        model = make_pipeline(PolynomialFeatures(degree), LinearRegression())
        predictions.append(model.fit(X, y).predict(x0)[0])
    print(degree, np.mean(predictions).round(3), np.var(predictions).round(3))
```

Bias and variance guide different interventions

Evidence	Likely issue	First experiment
High train and validation error	High bias	Richer features or model
Low train, high validation error	High variance	More data or stronger control
Both vary across groups	Distribution mismatch	Group specific audit
All scores unstable	Small evaluation sample	Repeated evaluation

Hoeffding concentration for one fixed rule

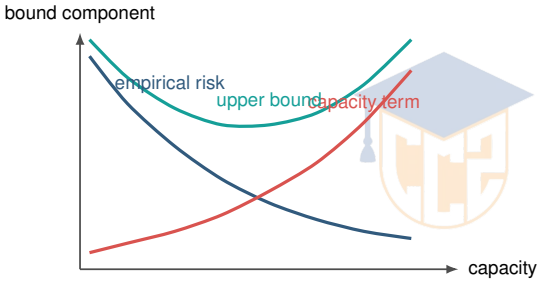
$$\Pr\left(|\widehat{R}_S(f) - R(f)| > \epsilon\right) \leq 2\exp(-2N\epsilon^2)$$

A finite hypothesis class needs a union bound

$$\Pr\left(\sup_{f\in\mathcal{H}} |\widehat{R}_S(f) - R(f)| > \epsilon\right) \leq 2|\mathcal{H}|e^{-2N\epsilon^2}$$

A generalization bound balances fit and capacity

$$R(\hat{f}) \lesssim \widehat{R}_S(\hat{f}) + \sqrt{\frac{\log|\mathcal{H}| + \log(1/\delta)}{N}}$$



Capacity counts label patterns rather than parameters alone

Question

How many distinct classifications can the family realize on points?

Reason

Different parameterizations can represent the same set of labelings.

Lesson

Parameter count is only a rough proxy for effective capacity.

Shattering requires every labeling of a point set

Definition

A set is shattered if every binary labeling can be realized by the hypothesis class.

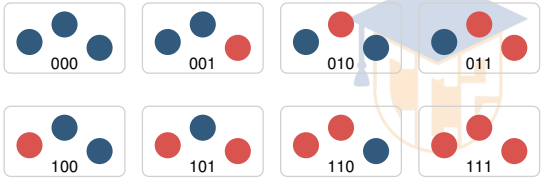
Three points

All eight label patterns must be possible.

One failure

Missing one pattern means that set is not shattered.

All labelings of three points



Threshold classifiers shatter one point but not two

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Family

$h_a(x) = \mathbf{1}(x \geq a)$ on the real line.

One point

Move the threshold to realize either label.

Two points

The left point cannot be positive while the right point is negative.

Linear separators have dimension dependent capacity

Affine separators

In d dimensions, the typical Vapnik-Chervonenkis dimension is $d + 1$.

Plane

Three noncollinear points can be shattered by lines.

Four points

Some arrangements and labelings cannot be separated.

Vapnik Chervonenkis dimension

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$$\text{VCdim}(\mathcal{H}) = \max\{m : \exists S,\ |S| = m,\ \mathcal{H} \text{ shatters } S\}$$

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A capacity bound shrinks with more data

$$R(\hat{f}) \lesssim \widehat{R}_S(\hat{f}) + O\left(\sqrt{\frac{h \log N + \log(1/\delta)}{N}}\right)$$

Structural risk minimization compares nested spaces

$$\mathcal{H}_1 \subset \mathcal{H}_2 \subset \cdots$$

Limits of generalization bounds

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Strength

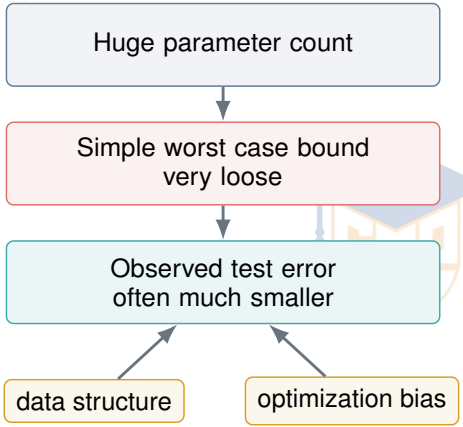
Bounds expose sample size, confidence, and capacity assumptions.

Limitation

Worst case constants can make numerical values conservative.

Modern puzzle

Large neural networks can generalize beyond simple parameter count predictions.



Theory identifies assumptions; experiments measure the actual system.

Check a loan model before using it

1. Define the future applicant population and asymmetric loss.
2. Split by application time and keep repeat applicants together.
3. Select features and penalty strength inside validation folds.
4. Report test loss, group differences, and uncertainty.
5. State which distribution changes would invalidate the estimate.

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Explain a generalization result together with its assumptions

- What future population does the result concern?
- How were the examples sampled and split?
- Which model family, loss, and selection procedure were used?
- What uncertainty or assumption limits the conclusion?

Statistical learning exam case

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Evidence

Training loss is near zero, validation loss varies widely across groups, and a random split looks excellent.

Task

Identify the likely failure, redesign evaluation, and propose one capacity experiment.

Strong answer

Connect each action to a specific observation and assumption.

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Learning theory self-check

1. Explain why low empirical risk does not guarantee low expected risk.
2. Design a valid split for one grouped or time ordered dataset.
3. Describe one regularization tradeoff using bias and variance.
4. State what must be shown to prove that a set is shattered.